Preview: y2002_q24 (v1)

Next number with a digit property: change the rightmost digits as little as possible

Part A: Each question is worth 3 points

1. The odometer of a car shows 7219 km. All the digits of this number are different.
After how many more kilometres will all the digits be different again?

A. 20 ***B.*** 11 ***C.*** 10 ***D.*** 9 ***E.*** 1

2. The odometer of a car shows 837549 km. All the digits of this number are different.
After how many more kilometres will all the digits be different again?

A. 1 ***B.*** 11 ***C.*** 13 ***D.*** 20 ***E.*** 21

3. The odometer of a bike shows 626 km. This number reads the same forwards and backwards.
After how many more kilometres will this happen again?

A. 1 ***B.*** 8 ***C.*** 9 ***D.*** 10 ***E.*** 110

4. The odometer of a bike shows 56765 km. This number reads the same forwards and backwards.
After how many more kilometres will this happen again?

A. 90 ***B.*** 100 ***C.*** 109 ***D.*** 110 ***E.*** 310

Part B: Each question is worth 4 points

5. The odometer of a car shows 6643 km. Not all of its digits are different.
How many kilometres ago did the odometer last show a number with all different digits?

A. 45 ***B.*** 44 ***C.*** 42 ***D.*** 35 ***E.*** 1

6. The odometer of a car shows 576789 km. Not all of its digits are different.
How many kilometres ago did the odometer last show a number with all different digits?

A. 341 ***B.*** 331 ***C.*** 301 ***D.*** 292 ***E.*** 291

7. A counter shows 569.
How much must it count up until it shows a number with all 3 digits the same?

A. 97 ***B.*** 98 ***C.*** 105 ***D.*** 110 ***E.*** 208

8. A counter shows 9925.
How much must it count up until it shows a number with all 4 digits the same?

A. 4 ***B.*** 59 ***C.*** 66 ***D.*** 73 ***E.*** 74

Part C: Each question is worth 5 points

9. The odometer of a scooter shows 689 km. Each digit is bigger than the digit before it.
After how many more kilometres will this happen again?

A. 99 ***B.*** 100 ***C.*** 101 ***D.*** 109 ***E.*** 110

10. The odometer of a scooter shows 2579 km. Each digit is bigger than the digit before it. After how many more kilometres will this happen again?

A. 10 ***B.*** 9 ***C.*** 8 ***D.*** 7 ***E.*** 1

Answer Key

1. **B**	2. **B**	3. **D**	4. **B**	5. **A**	6. **E**
7. **A**	8. **E**	9. **B**	10. **A**		

Solutions

1. B (11) — Counting up, every number before 7230 repeats a digit. 7230 has all different digits: 7230 – 7219 = 11. [2002 Q24 · all_different]

2. B (11) — Counting up, every number before 837560 repeats a digit. 837560 has all different digits: 837560 – 837549 = 11. [2002 Q24 · all_different]

3. D (10) — The next such number is 636, so 636 – 626 = 10. [2002 Q24 · next_palindrome]

4. B (100) — The next such number is 56865, so 56865 – 56765 = 100. [2002 Q24 · next_palindrome]

5. A (45) — Counting down from 6643, the first number with all different digits is 6598: 6643 – 6598 = 45. [2002 Q24 · last_time]

6. E (291) — Counting down from 576789, the first number with all different digits is 576498: 576789 – 576498 = 291. [2002 Q24 · last_time]

7. A (97) — The numbers with all digits the same are 555, 666, ... The next one after 569 is 666: 666 – 569 = 97. [2002 Q24 · same_digits]

8. E (74) — The numbers with all digits the same are 8888, 9999, ... The next one after 9925 is 9999: 9999 – 9925 = 74. [2002 Q24 · same_digits]

9. B (100) — Keep as many digits on the left as possible. The next such number is 789: 789 – 689 = 100. [2002 Q24 · going_up]

10. A (10) — Keep as many digits on the left as possible. The next such number is 2589: 2589 – 2579 = 10. [2002 Q24 · going_up]